STEVE NGAI

Senior .NET Developer

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PROFESSIONAL SUMMARY

Senior .NET Software Engineer with over 5+ years of experience building enterprise applications using C#, .NET, and Azure. Proven track record of delivering mission-critical solutions across banking, media, and digital services. Passionate about driving engineering excellence through innovation, process improvement, and cloud technologies. Led enterprise AI proof-of-concepts and modernization initiatives.

TECHNICAL CORE COMPETENCIES

Languages & Frameworks:	C# (.NET 10 / .NET 8 / .NET Framework 4.8), ASP.NET Core, Web API, Entity Framework Core, Razor Pages, Python, JavaScript, Vue.js, Blazor
Cloud & DevOps:	Azure (App Services, Container Apps, Functions, Key Vault, Service Bus, AKS), GitHub Actions, Azure DevOps CI/CD, Docker, Kubernetes, SonarQube
Generative AI & Data Science:	RAG Pipelines, Azure OpenAI, Azure AI Search, Azure Document Intelligence, Video Indexer, Vector DBs (ChromaDB, Pinecone), MLflow, Azure ML, Python (Pandas, LangChain)
Databases & Messaging:	SQL Server, PostgreSQL, MySQL, Elasticsearch, Azure Service Bus, Hangfire
Architecture & Security:	Clean Architecture, Vertical slice architecture, Microservices, RESTful APIs, SOLID, DRY, Design Patterns (Factory, Strategy, Repository), JWT

PROFESSIONAL EXPERIENCE

Senior .NET Developer

Arvato Systems Malaysia (MNC) | Nov 2023 – Present

Key Responsibilities:

- Contribute to the full software development lifecycle, including requirements analysis, feature design and implementation, code reviews, automated testing, debugging, performance optimization, and production support.
- Develop and maintain enterprise applications using C#, .NET, ASP.NET Core, Azure, SQL Server, and SharePoint, delivering scalable and maintainable solutions.
- Collaborate with cross-functional teams in an Agile Scrum environment, participating in sprint planning, daily stand-ups, reviews, and retrospectives while producing technical documentation such as database schemas, sequence diagrams, and system flow diagrams.

Key Initiatives, Innovation & Deliverables

- **AI Leadership & Certification:** Led a 5-member team to build an AI Onboarding Platform for 6 months. Appointed business unit AI Champion (2026). Drove team-wide AI adoption by engineering standardized github context instruction files, system rules, and agent skills to optimize Copilot accuracy and token efficiency. Shared weekly Copilot knowledge-sharing sessions for 16 engineers, monthly tech & security talk. Competed in the Arvato Systems Malaysia: AI Hackathon 2024 and earned the Udacity AI Product Manager Nanodegree (2026).
- **AI & MLOps POCs:** Developed a multimodal AI archival platform implementing RAG with Azure AI Search and Azure OpenAI using .NET 10 C# (2026), engineered an MLOps forecasting system tracking experiment drift via MLflow and Azure ML (2025), and architected and developed an AI Onboarding Platform using Blazor and and Vertical Slice Architecture (2026).
- **Internal Tooling & Observability:** Designed and built a self-hosted DevOps intelligence dashboard integrating GitHub REST APIs for real-time CI/CD visibility and operational automation across multi-repository microservice environments, featuring workflow health monitoring, branch propagation drift analysis, bulk workflow orchestration, and stale branch cleanup tooling.
- **CI/CD & DevOps Automation:** Reduced CI pipeline times by 66% via automated test-skipping and cut workflow execution times by 50% by migrating 10+ pipelines from Azure DevOps to GitHub Actions.
- **Security & Architecture:** Refactored the core security layer to resolve critical IDOR vulnerabilities, introduced secure data-masking policies for logging compliance, and advocated for modern design patterns (Strategy/Factory) and unit testing standards to elevate overall code quality.
- **Modernization & Load Testing:** Successfully migrated core enterprise applications from .NET 6 to .NET 8 and eliminated paid external testing tools (ie NBomber) by implementing a custom load testing engine using native Task parallelism, optimized HttpClient sockets, MSAL auth, and dynamic Bogus data streams for NotificationHub.
- **Governed Sales-Demo Platform:** Built an ASP.NET Core Razor Pages prototype supporting product catalogues, role-based approvals, template versioning, and customer-specific hosted demo publication. Implemented secure ZIP intake with manifest, dependency, token, path-traversal, hashing, quarantine, and promotion validation. Designed product adapter contracts with synthetic seed data, responsive bilingual UI, configurable branding, module selection, and approval workflows for Technical, Business, Sales, and Platform Admin personas.
- **Two POC on Terraform, Github Actions and JSM API:** POC 1 provisions a multi-tier containerized Azure stack—including App Service, SQL Database, Key Vault, and Storage—using Terraform, while leveraging Rover to visually map and preview the exact execution plan prior to provisioning (applying) the infrastructure. POC 2 focuses on GitOps governance and pipeline automation: it generates and preserves an exact Terraform plan, publishes the full plan summary to GitHub Pull Requests for review, and governs changes across GitHub Actions, Azure Functions, and Jira Service Management. In this pipeline, PipeGate creates and tracks the corresponding JSM ticket, while a separate manual-decision workflow accepts an approve or reject input before applying the exact saved plan file. Because the tested Jira request exposes no approval transitions, the current fallback stores the decision in Function memory and reports that Jira was unchanged, providing a clear demonstration baseline.

Microsoft .Net Developer

Quantum Storage Malaysia | Aug 2022 – Nov 2023

Key Initiatives, Innovation & Deliverables

- Automated Fulfilment: Developed a background microservice to reduce manual processing and improve operational efficiency.
- IT Asset Management: Built a CMDB system to improve IT asset tracking and management.
- Automated Testing: Introduced automated unit and integration testing to improve software quality and reliability.

- ETL Modernization & Cost Optimization: Developed a .NET 6 ETL service to migrate 40M records from MS Dataverse to SQL Server, reducing Azure costs by approximately $3,000/month.
- Performance Optimization: Upgraded the ETL solution from .NET 6 to .NET 8, achieving 16% faster bulk operations (7.3M records) and 30% faster smaller transactions (630K records), with no additional infrastructure cost.
- Process Automation: Built a static HTML email utility and Power Automate workflows to streamline internal processes and reduce repetitive manual tasks.
- Technology Innovation: Researched and evaluated SonarQube for code quality and Dagster for data orchestration to identify opportunities for future engineering improvements.

Full Stack Software Engineer
Affin Hwang Investment Bank | Aug 2020 – July 2022

Key Initiatives, Innovation & Deliverables
- Microservice Modernization: Replaced a legacy .NET Framework 4.x monolithic system with a scalable microservice-based automated data ingestion solution using .NET Core 3.1, improving scalability and operational efficiency.
- Performance Optimization: Optimized front-end and back-end services for eInvest GO and the Retail Online Platform, improving system responsiveness and reliability.
- Full-Stack Development: Expanded from backend to full-stack development by taking ownership of Vue.js frontend development, demonstrating adaptability and versatility across the technology stack.
- Process Improvement: Initiated and developed Jira productivity tracking reports to provide better visibility into team performance and delivery.
- System Reliability: Developed REST APIs and automated tests to improve application reliability and quality, contributing to a 20% increase in online trading account openings through eInvest GO.
- Consistent Performance: Consistently exceeded expectations in performance appraisals from August 2020 to December 2022.

SELECTED INNOVATION & PROOF-OF-CONCEPT (POC) PROJECTS

- Built a claims management POC using .NET 10 ASP.NET Core Minimal APIs, Angular, EF Core, SQL Server, Kafka, and .NET Aspire, implementing policy-first claim intake, claimant tracking, officer adjudication queues, manager liability reporting, audit history, optimistic concurrency, and validation-driven state transitions. Designed an event-driven workflow with a transactional outbox, retry and dead-letter handling, idempotent consumers, and an independent officer read-model projection, while organizing backend capabilities as vertical slices and integrating the Angular frontend through Aspire.
- **Azure MLOps Pipeline (2025):** Built an Azure ML and MLflow pipeline automating data prep, model training, tracking, and compute orchestration for automated model retraining.
- **Configurable Python RAG Pipeline (2025):** Developed a dynamic RAG pipeline using Sentence Transformers, ChromaDB, and LM Studio API (Zephyr / Llama 3.2) featuring configurable text chunking and similarity search.
- **Document Intelligence Assistants (2025):** Built Pinecone and BART Transformer-based AI assistants in Python/Streamlit enabling context-strict natural language QA and intelligent PDF summarization.
- **Cloud Native .NET Aspire Deployment (2024):** Provisioned 15 Container Apps in 11 minutes on Azure via Azure Developer CLI (azd) using .NET Aspire.
- **Asynchronous Event Messaging (2024):** Engineered an enterprise messaging architecture using Azure Functions, Azure Service Bus, and Hangfire for resilient asynchronous job scheduling.
- **Microservices & Kubernetes Orchestration:** Configured eShopOnContainers with Azure Kubernetes Service (AKS), Docker Desktop, Helm, and Azure Active Directory integration.

CAREER BACKGROUND & EXPERIENCE SUMMARY

With 17+ total years of technical experience across multi-national IT environments, the career breakdown encompasses:

- **Software Design & Engineering:** 5+ Years — Hands-on architecture, full-stack C#.NET, Web API, Entity Framework, and Azure deployment.
- **Application Support (Level 2 & Level 3):** 5 Years — Deep operational expertise maintaining mission-critical banking systems (including eSPICK for Bank Negara Malaysia) and high-availability enterprise services.
- **Software Engineering Instructor:** 5 Years — Mentoring and training emerging software engineers in modern development methodologies and OOP principles.
- **Technical Management & Leadership:** 2 Years — Technical leadership, quality management, and cross-functional Agile project oversight.

EDUCATION & CERTIFICATIONS

- **B.Sc. (Hons) in Computing:** Staffordshire University, UK | First Class / High Honors
- **Udacity AI Product Manager Nanodegree (2026):** Hands-on focus on PRD generation, product roadmap prioritization, RAG architectures, and responsible AI system design.
- **Key Honors & Recognition:** Appointed BU AI Champion (Arvato Systems, 2026) | Performance Excellence Review (2024)

DOMAIN KNOWLEDGE

- Commercial & Investment Banking, Central Banking Systems (eSPICK), Payment Gateways, Media & Broadcasting Technology, Property & Manufacturing Operations.